

# Talha Khan

Assistant Professor

Department of Information Sciences and Technologies  
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<https://talhakhan13.github.io/>  
[Google Scholar](#)

**Research Interests:** Human-Computer Interaction, Mixed Reality

## PROFESSIONAL EXPERIENCE

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| <ul style="list-style-type: none"><li>• <b>Assistant Professor</b><br/>College of Computing, Grand Valley State University, USA<ul style="list-style-type: none"><li>• Director, Immersive Interaction Lab</li></ul></li></ul> | AUG. 2025 - CURRENT   |
| <ul style="list-style-type: none"><li>• <b>Graduate Research Fellow</b><br/>Department of Computer Science, University of Pittsburgh, USA</li></ul>                                                                            | SEP. 2024 - APR. 2025 |
| <ul style="list-style-type: none"><li>• <b>Research Intern, Human-Centered AI</b><br/>Toyota Research Institute, USA</li></ul>                                                                                                 | MAY. 2024 - AUG. 2024 |
| <ul style="list-style-type: none"><li>• <b>Research Intern, Augmented Perception Lab</b><br/>Carnegie Mellon University, USA<br/>Mentor: <a href="#">DR. DAVID LINDLBAUER</a></li></ul>                                        | MAY. 2023 - AUG. 2023 |
| <ul style="list-style-type: none"><li>• <b>Graduate Student Researcher</b><br/>Department of Computer Science, University of Pittsburgh, USA</li></ul>                                                                         | MAY. 2021 - APR. 2024 |

## EDUCATION

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| <ul style="list-style-type: none"><li>• <b>PhD in COMPUTER SCIENCE</b><br/>University of Pittsburgh, USA<ul style="list-style-type: none"><li>• Thesis: Towards Integrating Mixed Reality Into Physical Workspaces</li><li>• Advisor: <a href="#">DR. JACOB BIEHL</a></li></ul></li></ul> | AUG. 2020 - APR. 2025 |
| <ul style="list-style-type: none"><li>• <b>MS in COMPUTER SCIENCE</b><br/>University of California, Irvine, USA</li></ul>                                                                                                                                                                 | SEP. 2018 - JUN. 2020 |
| <ul style="list-style-type: none"><li>• <b>BS in COMPUTER SCIENCE</b><br/>Lahore University of Management Sciences, Pakistan</li></ul>                                                                                                                                                    | AUG. 2014 - JUN. 2018 |

## TEACHING EXPERIENCE

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| <ul style="list-style-type: none"><li>• <b>Course Instructor</b><br/>Grand Valley State University, USA<br/>Courses: <i>Computer Science 1</i> (Introduction to Python), <i>AR/VR Design and Research</i></li></ul>                            | AUG. 2025 - CURRENT   |
| <ul style="list-style-type: none"><li>• <b>Graduate Teaching Assistant</b><br/>University of Pittsburgh, USA<br/>Courses: <i>Algorithm Implementation</i>, <i>Programming Language for Web Applications</i></li></ul>                          | SEP. 2020 - APR. 2021 |
| <ul style="list-style-type: none"><li>• <b>Graduate Teaching Assistant</b><br/>University of California, Irvine, USA<br/>Courses: <i>Intermediate Programming (Python)</i>, <i>Data Structures (C++)</i>, <i>Introduction to C++</i></li></ul> | OCT. 2018 - MAY. 2020 |
| <ul style="list-style-type: none"><li>• <b>Undergraduate Teaching Assistant</b><br/>Lahore University of Management Sciences, Pakistan<br/>Courses: <i>Applied Probability (Graduate-level)</i>, <i>Probability</i></li></ul>                  | SEP. 2017 - MAY. 2018 |

## PUBLICATIONS

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### Conference Publications & Proceedings

- [C.11] CommonGround: Externalizing Mixed Reality Interfaces for Active Bystander Collaboration  
Khadija Arif, **Talha Khan**  
*ACM CSCW2026*
- [C.10] ReadAR: Utilizing Mixed Reality for Physical Document Digitization  
Liam Adams, **Talha Khan**  
*ACM IMX 2026*
- [C.9] Utilizing VLMs for Identifying Child Safety Hazards  
**Talha Khan**, Ahmed Ashraff Butt  
*ACM CHI 2026*
- [C.8] MR-MDEs: Exploring the Integration of Mixed Reality into Multi-Display Environments  
Griffin Hurt, **Talha Khan**, Nicolás Kass, Anthony Tang, Edward Andrews, Jacob Biehl  
*PACM HCI (ACM ISS 2025)*
- [C.7] Shadows of Reality: Enhancing Bystander Awareness of Mixed Reality Interfaces  
**Talha Khan**, Abigail Zimmerman, Edward Andrews, David Lindlbauer, Jacob Biehl  
*ACM SUI 2025*
- [C.6] Don't Block My Stuff: Fostering Personal Object Awareness in Multi-user Mixed Reality Environments  
**Talha Khan**, David Lindlbauer  
*PACM HCI (ACM ISS 2024)*
- [C.5] Understanding Perceived Utility and Comfort of In-Home General-Purpose Sensing through Progressive Exposure  
Pranut Jain, Andrew Xu, Thomas Downes, Injung Kim, **Talha Khan**, Jacob Biehl, Adam Lee.  
*PACM HCI (ACM CSCW 2024)*
- [C.4] Endovascular: Utility of Mixed Reality to Segment Large Displays in Surgical Settings  
Griffin Hurt, **Talha Khan**, Michael Kann, Edward Andrews, Jacob Biehl  
*IEEE VRW 2024*
- [C.3] Future of Surgical Mixed Reality: Cutting-Edge or Cutting Too Deep?  
**Talha Khan**, Edward Andrews, Jacob Biehl  
*IEEE VRW 2024*
- [C.2] Understanding Effects of Visual Feedback Delay in AR on Fine Motor Surgical Tasks  
**Talha Khan**, Toby Zhu, Thomas Downes, Lucille Cheng, Nicolás Kass, Edward Andrews, Jacob Biehl.  
*IEEE TVCG (ISMAR 2023) Best Paper Award*
- [C.1] AR in the OR: exploring use of augmented reality to support endoscopic surgery  
**Talha Khan**, Edward Andrews, Paul Gardner, Arka Mallela, Jeffrey Head, Joseph Maroon, Georgios Zenonos, Dmitriy Babichenko, Jacob Biehl  
*ACM IMX 2022*

### Journal Publications

- [J.5] Application of Mixed Reality in Endoscopic Streaming during Skull Base Surgery  
Yujin Choi, Shovan Bhatia, Adam Monek, Harrison Alexander, **Talha Khan**, Brendan Szczepkowski, Griffin Hurt, Anthony Tang, Michael Kann, Regan Shanahan, Brock Gjesdal, Rishi Basdeo, Kyle Affolter, David Fernandes Cabral, Georgios Zenonos, Paul Gardner, Jacob Biehl, Edward Andrews  
*Journal of Neurological Surgery Part B: Skull Base 2026 (Abstract)*  
*Operative Neurosurgery 2026 (Full Paper)*
- [J.4] Evolution of the meta-neurosurgeon: A systematic review of the current technical capabilities, limitations, and applications of augmented reality in neurosurgery  
Nikhil Sharma, Arka Mallela, **Talha Khan**, Stephen Paul Canton, Nicolás Kass, Fritz Steuer, Jacquelyn Jardini, Jacob Biehl, Edward Andrews.  
*Surgical Neurology International 2024*
- [J.3] Solar-powered Parking Analytics System using Deep Reinforcement Learning  
Yoonas Rezai, **Talha Khan**, Stephen Lee, Daniel Mossé  
*ACM TOSN 2023*

- [J.2] A systematic comparison of the accuracy of monocular RGB tracking and LiDAR for neuronavigation  
**Talha Khan**, Jacob Biehl, Edward Andrews, Dmitriy Babichenko  
*Healthcare Technology Letters* 2022
- [J.1] Streaming 2D-Endoscopic Video into an Augmented Reality Headset Display: A Feasibility Study  
Edward Andrews, **Talha Khan**, Arka Mallela, Joseph Maroon, Jacob Biehl, David Fernandes-Cabral, Zachary Gersey,  
Hussam Abou-Al-Shaar, Paul Gardner, Georgios Zenonos  
*Journal of Neurological Surgery Part B: Skull Base* 2022

## MENTORSHIP EXPERIENCE

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- Khadia Arif JAN. 2026 - CURRENT  
MS Human-Centered Computing, Grand Valley State University, USA  
**Project:** *ProjectAR: Externalizing Mixed Reality Interfaces to Support Bystander Collaboration*
- Liam Adams AUG. 2025 - CURRENT  
MS Human-Centered Computing, Grand Valley State University, USA  
**Project:** *ReadAR: Utilizing Mixed Reality for Physical Document Digitization*
- Abigail Zimmerman SEP. 2024 - JAN. 2025  
BS Digital Narrative and Interactive Design, University of Pittsburgh, USA  
**Project:** *Shadows of Reality: Enhancing Bystander Awareness of Mixed Reality Interfaces (ACM SUI 2025)*
- Bo-Chen Kuo JAN. 2024 - APR. 2024  
MS Computer Science, University of Pittsburgh, USA  
**Project:** *Automatic Segmentation of Medical Appliance Buttons*
- Elliott Finney JAN. 2024 - APR. 2024  
BS Computer Science, University of Pittsburgh, USA  
**Project:** *Multi-modal Mixed Reality Interfaces for Medical Appliances*
- Griffin Hurt MAY. 2023 - APR. 2024  
BS Computer Science, University of Pittsburgh, USA  
**Project:** *MR-MDEs: Exploring the Integration of Mixed Reality into Multi-Display Environments (ACM ISS 2025)*
- Chloe Dahan JUN. 2021 - AUG. 2021  
BS Digital Narrative and Interactive Design, University of Pittsburgh  
**Project:** *Automatic Human-Humanoid Gesture Imitation*
- Aiden Dorneich JUN. 2021 - AUG. 2021  
Research Intern Facet Lab, University of Pittsburgh, USA  
**Project:** *Automatic Human-Humanoid Gesture Imitation*

## HONORS AND AWARDS

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- CS 50 Fellowship Award, Department of Computer Science, University of Pittsburgh APR. 2024
- Best Journal Paper Award, IEEE International Symposium on Mixed and Augmented Reality OCT. 2023
- Honorable Mention Award, Link Foundation Modelling, Simulation & Training Fellowship JUN. 2022

## MEDIA APPEARANCES

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- **Pitt To The Point** MAR. 2024  
[Revolutionizing Healthcare With Mixed Reality](#)

## INVITED TALKS

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- **Guest Lecture, Arizona State University** FEB. 2025  
*Impact of Visual Delays in Mixed Reality on Surgical Performance*

- **Dean's Spotlight Series, University of Pittsburgh**  
*AR in the OR: An Interdisciplinary Journey Towards the Holographic Operation Room*

JAN. 2023

## LEADERSHIP EXPERIENCE

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- **Business Manager** MAR. 2024 - APR. 2025  
Doctoral Guild at Pitt, University of Pittsburgh, USA
- **Founding Member** APR. 2023 - CURRENT  
[Surreality Lab](#), University of Pittsburgh, USA

## REVIEWING SERVICE

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- ACM International Conference on Interactive Surfaces and Spaces (ACM ISS 2026)
- Springer Nature Virtual Reality (2026)
- IEEE International Conference on Virtual Reality and 3D User Interfaces (*IEEE VR 2026*)
- ACM CHI Conference on Human Factors in Computing Systems (*ACM CHI 2026*)
- ACM Symposium on Spatial User Interaction (*ACM SUI 2025*)
- ACM CHI Conference on Human Factors in Computing Systems (*ACM CHI 2025*)
- ACM Symposium on User Interface Software and Technology (*ACM UIST 2024*)
- ACM CHI Conference on Human Factors in Computing Systems (*ACM CHI 2024*)
- ACM CHI Conference on Human Factors in Computing Systems (*ACM CHI 2023*)
- ACM Conference On Computer-Supported Cooperative Work And Social Computing (*ACM CSCW 2023*)

## VOLUNTEER SERVICE

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- Student Volunteer, ACM Conference on Interactive Surfaces and Spaces (ACM ISS 2023)
- Student Volunteer, ACM Conference on Human Factors in Computing Systems (ACM CHI 2022)

## REFERENCES

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### DR. JACOB BIEHL

Associate Professor, Department of Computer Science, University of Pittsburgh, USA

🌐 <https://jtbiehl.github.io/> | ✉ [biehl@pitt.edu](mailto:biehl@pitt.edu)

### DR. DAVID LINDLBAUER

Assistant Professor, Human-Computer Interaction Institute, Carnegie Mellon University, USA

🌐 <https://www.davidlindlbauer.com/> | ✉ [davidlindlbauer@cmu.edu](mailto:davidlindlbauer@cmu.edu)

### DR. ERIN WALKER

Professor, Department of Computer Science, University of Pittsburgh, USA

🌐 <https://erinwalker.academicwebsite.com/> | ✉ [eawalker@pitt.edu](mailto:eawalker@pitt.edu)

### DR. EDWARD ANDREWS

Assistant Professor, Department of Neurological Surgery, University of Pittsburgh, USA

✉ [andrewse2@upmc.edu](mailto:andrewse2@upmc.edu)